

RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

3. RF and M/W Network Theory and Analysis

4 hrs · asked in **18 of 24 papers** · 2 syllabus sub-topics · **193 marks, 10.1% of the archive** → definitions, one derivation, four matrix properties and one device (the magic tee) that answers half the chapter.

READ THIS FIRST

Chapter 3 is the cheapest 8 to 10 marks in the paper, because the same three asks recur:

1. **Why S-parameters, and how do they differ from h / Z / Y / ABCD?** (13 papers) → one reason list plus one comparison table, §3.1.
2. **Derive the S-parameters of a two-port network** (7 papers) → the same six lines every time, §3.2. Learn it as a script.
3. **Identify the passive device from this S-matrix** (5 papers) → **in every one of the five it is the magic tee**, in four different disguises. Learn the fingerprint in §3.7 and the matrix in §3.6.

Two more traps worth knowing before the paper starts:

- The radar questions (“two transmitters, twice the power”, “half the power to the antenna via a duplexer”, 4 papers) are the **same magic tee** asked as a design problem, §3.8. Nothing new has to be learnt for them.
 - An examiner who writes “3-port network” almost always wants the **impossibility theorem** (§3.5), not just a 3×3 matrix. Stating that a 3-port cannot be lossless, reciprocal and matched at the same time, and then naming the three escapes, is what separates a 4 from an 8.
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3.1 Why S-parameters: the Limits of Z, Y, h and ABCD

Why are S-parameters used in microwave network analysis? Compare S-parameters with h-parameters

2 3 4 4+4 5+5

TOP 13/24 69 Bh, 70 Bh, 70 Ma, 73 Ma, 73 Bh, 74 Ma, 75 Bh, 78 Ch, 79 Bh, 80 Ba, 80 Bh, 80 Ch, 81 Bh · the single most repeated ask in the chapter

What the low-frequency parameter sets need

- All four classical sets relate **total voltage and total current** at the terminals:
 - $[V] = [Z][I]$, $Z_{ij} = V_i/I_j$ with all other ports **open**, $I_k = 0$.
 - $[I] = [Y][V]$, $Y_{ij} = I_i/V_j$ with all other ports **shorted**, $V_k = 0$.
 - h-parameters: mixed, h_{11} and h_{21} need a **short** on port 2, h_{12} and h_{22} need an **open** on port 1.
 - ABCD: cascade form, still defined on V and I .
- Two things are therefore assumed:
 - V and I are **unique** at a terminal pair (true only when $d \ll \lambda$).
 - Ideal **open** and **short** terminations are available.
- Both assumptions fail above roughly 1 GHz.

The five reasons they break at microwave

1. **No unique V and I :**
 - Circuit size is comparable to λ , so voltage varies along the conductor.
 - In a waveguide there is no unique pair of conductors at all, so "the voltage" has no single definition.
2. **Open and short cannot be realised:**
 - A physical short has series lead inductance, an open has fringing capacitance.
 - A short is a short at **one frequency and one reference plane** only: $\lambda/4$ away it looks like an open.
3. **Total reflection is destructive:**
 - $| \Gamma | = 1$ sends all the power back into the device.
 - Transistors and negative-resistance diodes **oscillate or burn out** under open or short test conditions.
4. **No instrument measures RF V and I :**
 - Probes load the circuit and cannot follow GHz waveforms.
 - **Power** is what a microwave detector actually reads.
5. **Bias shifts:**
 - Open and short terminations change the DC operating point and the junction capacitance of active devices, so the measurement is not of the device in use.

What S-parameters do instead

- Defined on **travelling waves**, not on totals:
 - a_i = wave incident on port i , b_i = wave leaving port i .
 - Normalised so $|a_i|^2$ and $|b_i|^2$ are **powers**, in watts.
- Every port is terminated in Z_0 , a **matched load**:
 - Easy to build and broadband, unlike open or short.
 - No reflection back into the device $\rightarrow$ no oscillation, no damage, bias undisturbed.
- Measured with **directional couplers** plus a vector network analyser, which separate incident from reflected wave and read magnitude and phase directly.
- Every practical figure of merit falls straight out: gain, return loss, insertion loss, isolation, VSWR, stability (§3.4).
- Valid at **any** frequency, so the same formalism covers DC to mmWave.
- Cascadable after conversion to the transfer (T) form, so a chain of measured blocks can be multiplied out.

h-parameters vs S-parameters

Aspect	h-parameters	S-parameters
Variables	Total V, I	Incident, reflected waves a, b
Termination	Short and open	Matched Z_0
Measured	Voltage, current	Power (magnitude + phase)
Instrument	Curve tracer, bridge	Vector network analyser
On active devices	Oscillation, damage	Safe, bias preserved
Frequency	Below ≈ 100 MHz	Any, chosen for RF / M/W
Broadband	No, termination drifts	Yes
Cascading	Awkward	Via T-matrix

Answer skeleton for the 4-mark version 4

- One line on what fails: no unique V / I , open and short unrealisable, active devices damaged, no RF voltmeter.
- One line on the fix: normalised travelling waves measured against matched Z_0 loads with directional couplers.
- One line on the payoff: gain, RL, IL, VSWR and stability read straight off $[S]$.
- Close with $[b] = [S][a]$ and the two-port matrix. Four sentences, full marks.

3.2 Definition and Derivation for a Two-Port Network

Using a two-port network, define and derive the S-parameters 4 5 6 10

TOP 7/24 69 Bh, 70 Bh, 70 Ma, 78 Ch, 80 Ba, 80 Bh, 80 Ch · write it as a script: figure, normalisation, matrix, four conditions

Step 1: the wave variables

- At port i the line carries a forward wave V_i^+ and a backward wave V_i^- . Normalise both by $\sqrt{Z_0}$:

$$a_i = \frac{V_i^+}{\sqrt{Z_0}}, \quad b_i = \frac{V_i^-}{\sqrt{Z_0}}$$

- The normalisation is chosen so that the squared magnitudes are powers:

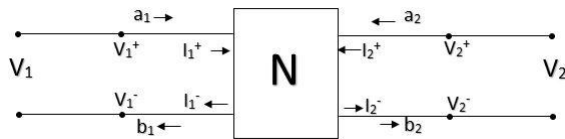
$$P_i^{\text{inc}} = \frac{1}{2}|a_i|^2, \quad P_i^{\text{ref}} = \frac{1}{2}|b_i|^2$$

(the $\frac{1}{2}$ disappears if a, b are taken as rms).

- In terms of the total quantities:

$$V_i = \sqrt{Z_0}(a_i + b_i), \quad I_i = \frac{a_i - b_i}{\sqrt{Z_0}}$$

so nothing is lost, only re-expressed.



Port 1 and port 2 of a general two-port, each carrying an incident wave a and an emerging wave b . Source: RF Pulchowk Ch3 deck.

Step 2: the defining matrix

- The network is linear, so each outgoing wave is a weighted sum of all the incoming ones:

$$b_1 = S_{11}a_1 + S_{12}a_2, \quad b_2 = S_{21}a_1 + S_{22}a_2$$

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \quad \text{i.e.} \quad [b] = [S][a]$$

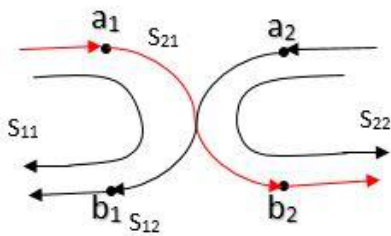
- Subscript order:** in S_{ij} the **second** index j is the port driven, the **first** index i is the port observed. S_{21} is “out of 2, in at 1”.
- Written out, the matrix is the complete input-output description of the black box: knowing four numbers replaces solving for every internal voltage and current.

Step 3: isolate each parameter

- Terminate the unused port in Z_0 . A matched load reflects nothing, so its incident wave vanishes and one column of the matrix survives at a time.

Parameter	Condition	Meaning
$S_{11} = \frac{b_1}{a_1}$	$a_2 = 0$	Input reflection coefficient, port 2 matched
$S_{21} = \frac{b_2}{a_1}$	$a_2 = 0$	Forward transmission (gain or insertion loss)
$S_{12} = \frac{b_1}{a_2}$	$a_1 = 0$	Reverse transmission (isolation, feedback)
$S_{22} = \frac{b_2}{a_2}$	$a_1 = 0$	Output reflection coefficient, port 1 matched

- $a_2 = 0$ means “**port 2 terminated in Z_0 ”, never “port 2 open”**. Writing “open circuit” here loses the mark, because it is exactly the condition S-parameters were invented to avoid.



Signal-flow graph of the same two-port: four branches, one per S-parameter. Useful when the question asks for a “flow diagram”.

Step 4: tie S_{11} back to impedance

- With port 2 matched, port 1 simply sees the input impedance Z_{in} , so

$$S_{11} = \Gamma_{in} = \frac{Z_{in} - Z_0}{Z_{in} + Z_0}$$

- This is the bridge to Chapter 2: a one-port load has the single entry $[S] = [\Gamma_L]$, and a perfectly matched two-port has $[S] = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (§2.8).

Step 5: generalise to n ports

- $b_i = \sum_{j=1}^n S_{ij} a_j$, an $n \times n$ matrix with n^2 entries:

$$S_{ij} = \left. \frac{b_i}{a_j} \right|_{a_k=0, k \neq j}$$

- Rows and columns both equal the port count, so a 3-port gives 3×3 and a magic tee gives 4×4 .
- **Reference impedance:** every entry is defined against the same Z_0 (usually 50Ω), so a quoted matrix without its Z_0 and its frequency is incomplete.

3.3 Properties of the S-Matrix

State and justify the properties of the scattering matrix 4 4+4 5 6

TOP 8/24 73 Bh, 74 Bh, 75 Bh, 78 Ch, 79 Bh, 80 Bh, 81 Bh, 82 Bh · these four tests are also the tool that identifies a device from its matrix

P1. Zero diagonal $\Leftrightarrow$ matched port

- $S_{ii} = 0$ means a wave arriving at port i is not reflected, so that port is matched to Z_0 .
- “Matched at all ports” means the whole diagonal is zero.

P2. Symmetry $\Leftrightarrow$ reciprocity

- $S_{ij} = S_{ji}$ for all i, j , i.e. $[S] = [S]^T$.
- **Holds** for networks built of passive, linear, isotropic media: lines, stubs, tees, couplers, resistors, capacitors, transformers.
- **Fails** for magnetised ferrite (circulator, isolator, gyrator) and for active devices (amplifier, transistor), where $S_{21} \neq S_{12}$ by design.
- So a matrix that is **not** symmetric can never be a passive isotropic device.

P3. Unitary $\Leftrightarrow$ lossless

- A lossless network redirects power but never dissipates it, which forces

$$[S]^* [S] = [I]$$

- Reading that out column by column gives the two tests actually used in exams:
 - **Column-sum test** (each column carries unit power):

$$\sum_{k=1}^n |S_{ki}|^2 = 1 \quad \text{for every } i$$

- **Orthogonality test** (distinct columns are orthogonal):

$$\sum_{k=1}^n S_{ki} S_{kj}^* = 0 \quad \text{for every } i \neq j$$

- If a column sums to less than 1, the shortfall is the fraction of incident power the network **dissipates**, which is how “is it lossless?” is answered numerically (§3.4 and Problem 1 of the companion).

P4. Shifting the reference plane multiplies by a phase

- Move the reference plane of port i outward by an electrical length $\theta_i = \beta \ell_i$. Then

$$S'_{ij} = S_{ij} e^{-j(\theta_i + \theta_j)}$$

- Diagonal terms pick up $e^{-2j\theta_i}$, because the wave travels the added length twice.
- **Magnitudes are unchanged:** $|S'_{ij}| = |S_{ij}|$. Return loss, insertion loss and VSWR do not depend on where the plane is drawn; only phase does.
- This is why an S-parameter measurement is meaningless until the planes are declared, and why a VNA must be calibrated to a stated plane.

P5. Complex, frequency dependent, n^2 entries

- Each S_{ij} carries a magnitude and an angle, and both move with frequency, so a quoted set is meaningless without its f and its Z_0 .
- An n -port has n^2 coefficients; symmetry and structural zeros cut the number of independent ones sharply (the magic tee has 16 entries but only 2 distinct values).

P6. Symmetric and antimetric two-ports

- **Symmetric:** reciprocal and $S_{11} = S_{22}$. The network looks the same from either end.
- **Antimetric:** $S_{11} = -S_{22}$.

The four tests as an exam checklist

Test	Look for	Conclusion
Diagonal	$S_{ii} = 0?$	That port is matched
Transpose	$[S] = [S]^T?$	Reciprocal (passive) or not (ferrite / active)
Column sums	$\sum_k S_{ki} ^2 = 1?$	Lossless, else lossy by the shortfall
Column dots	$\sum_k S_{ki} S_{kj}^* = 0?$	Confirms lossless, ties the phases

Run the four in this order on any matrix the examiner prints: they are enough to name the device (§3.7) and to answer every “judge the condition” rider.

3.4 Return Loss, Insertion Loss and the Other Figures of Merit

Define return loss and insertion loss, and the losses associated with a two-port network 2 1 4

HOT 3/24 72 Ash, 79 Bb, 82 Bb · small marks, but they are the language every other answer is written in

The three powers at a port

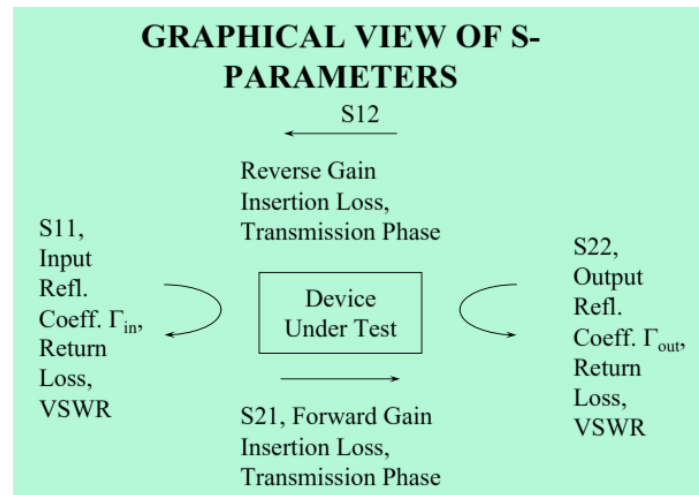
- P_i incident, P_r reflected, P_t transmitted; whatever is left over is dissipated.
- Everything below is a ratio of two of those, expressed in dB and rewritten in terms of $[S]$.

Quantity	In terms of $[S]$	What it tells you
Return loss	$RL = -20 \log_{10} S_{11} $	How well port 1 is matched. Bigger is better
Insertion loss	$IL = -20 \log_{10} S_{21} $	Power lost passing through. 0 dB is ideal
Isolation	$-20 \log_{10} S_{12} $	Leakage backwards. Bigger is better
Forward gain	$G = 20 \log_{10} S_{21} $	Same number as IL , opposite sign, used for amplifiers
VSWR	$\frac{1 + S_{11} }{1 - S_{11} }$	The same match on a 1 to ∞ scale
Reflection (mismatch) loss	$-10 \log_{10} (1 - S_{11} ^2)$	The part of P_i that never enters the network
Dissipated fraction	$1 - \sum_k S_{k1} ^2$	What the network eats. Zero for a lossless network

- **Useful anchors:** $|S_{11}| = 0.5 \rightarrow RL = 6 \text{ dB}$, $VSWR = 3$; $|S_{11}| = 0.1 \rightarrow RL = 20 \text{ dB}$, $VSWR = 1.22$; $|S_{11}| = 0 \rightarrow$ perfect match, $RL = \infty$.

Reading a device off its numbers

- Small $|S_{11}|$, $|S_{22}| \rightarrow$ well matched at both ends.
- $|S_{21}| > 1 \rightarrow$ **active**: the device has gain.
- $|S_{21}| < 1$ with $S_{12} = S_{21} \rightarrow$ **passive and reciprocal**: an attenuator, filter or line.
- $|S_{12}| \ll |S_{21}| \rightarrow$ **unilateral**: an amplifier or isolator; the smaller $|S_{12}|$ is, the more stable the amplifier tends to be.
- This is the whole method for 82 Bb, which prints two amplifier S-parameter sets and asks only for “the basic differences”. Problem 2 of the companion works it.



What each S-parameter of a device under test is called in the lab. Source: RF Pulchowk Ch3 deck.

Sign conventions that cost marks

- Return loss and insertion loss are quoted as **positive** dB for a passive device. The minus sign is already in the definition.
- 20 log for a **ratio of waves** (S is an amplitude ratio), 10 log for a **ratio of powers**. Mixing the two is the commonest slip.

verified/added Correction to the source notes

- The RF Pulchowk deck prints $IL \text{ (dB)} = 10 \log (|a_1|^2 / |b_1|^2)$.
- $|b_1|$ is the wave coming **back out of port 1**, so that expression is the return loss, not the insertion loss.
- Insertion loss compares the incident wave with what leaves the **far** port:

$$IL = 10 \log \frac{|a_1|^2}{|b_2|^2} = 20 \log \frac{1}{|S_{21}|}$$

which is what the deck's own next line prints. Use the $|S_{21}|$ form.

3.5 Three-Port Networks and Why the Ideal One Cannot Exist

Analyse a three-port network using S-parameters. Write the properties of a 3-port network and define its characteristic parameters **4 4+4 2+6 5+2**

HOT 6/24 72 Ash, 73 Ma, 73 Bh, 74 Ma, 80 Bh, 82 Bh · the theorem is the answer, the matrix alone is half marks

The general 3-port

- Nine coefficients:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

- **Characteristic parameters** an examiner expects you to define:
 - **Matching:** S_{ii} , hence return loss and VSWR at each port.
 - **Coupling / division ratio:** $|S_{ij}|^2$, the fraction of the input power delivered to port i .
 - **Isolation:** $-20 \log |S_{ij}|$ for the pair meant to be decoupled.
 - **Phase relation:** $\arg S_{13}$ against $\arg S_{23}$, i.e. whether the two output arms are in phase or in antiphase.
 - **Insertion loss** and **bandwidth** over which all of the above hold.

Theorem: no 3-port is lossless, reciprocal and matched at once

- **Assume all three.** Matched gives $S_{11} = S_{22} = S_{33} = 0$, reciprocal gives $S_{ij} = S_{ji}$, so

$$[S] = \begin{bmatrix} 0 & S_{12} & S_{13} \\ S_{12} & 0 & S_{23} \\ S_{13} & S_{23} & 0 \end{bmatrix}$$

- **Apply the orthogonality test** to each pair of columns:

$$S_{13}^* S_{23} = 0, \quad S_{12}^* S_{23} = 0, \quad S_{12}^* S_{13} = 0$$

so at least **two** of S_{12}, S_{13}, S_{23} must be zero.

- **Apply the column-sum test:**

$$|S_{12}|^2 + |S_{13}|^2 = 1, \quad |S_{12}|^2 + |S_{23}|^2 = 1, \quad |S_{13}|^2 + |S_{23}|^2 = 1$$

which requires all three to be non-zero.

- The two demands **contradict**, so no such network exists.
Give up exactly one of the three properties.

The three escapes, and the device each one gives

Give up	Device	Consequence
Lossless	Resistive divider, matched 3-port	All ports matched and reciprocal, but power is burnt in the resistors
Reciprocal	Circulator (ferrite)	Matched and lossless, but $S_{ij} \neq S_{ji}$; power circulates $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$
Matched at every port	E-plane tee, H-plane tee, T-junction	Lossless and reciprocal, but only the side arm can be matched

- **Ideal circulator**, matched and lossless but not reciprocal:

$$[S] = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

Terminate port 3 in a matched load and it becomes an **isolator**.

- A **4-port** escapes the trap entirely: a directional coupler and a magic tee are matched, lossless *and* reciprocal all at once, which is exactly why the exam favourite is a 4-port (§3.6).

E-plane tee: the worked 3-port (§3.6 has the full matrix)

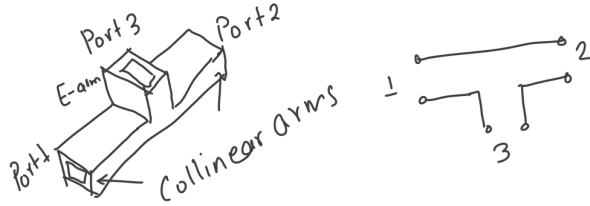
- Ports 1 and 2 collinear, port 3 the E-arm, assumed matched to the junction.
- Three physical facts start the derivation:
 - An input at the E-arm splits into two outputs that are **180° out of phase**: $S_{23} = -S_{13}$.
 - Port 3 is matched: $S_{33} = 0$.
 - The junction is passive and isotropic: $S_{ij} = S_{ji}$.
- Four unknowns are left, and the unitary property fixes all four (§3.6, and Problem 5 of the companion works every line).
- **Ports 1 and 2 cannot be matched**, and indeed the answer gives $S_{11} = S_{22} = 1/2$, which is the theorem showing up in the arithmetic.

3.6 S-Matrix Catalogue of the Standard Devices

Prepare the S-matrices of a perfectly working E-plane Tee, H-plane Tee and a Duplexer, and explore their uses 4+4 8 3+7

HOT 4/24 78 Ch, 81 Ba, 80 Bh, 82 Bh · memorise these five matrices and half the chapter is already answered

E-plane tee (series tee)

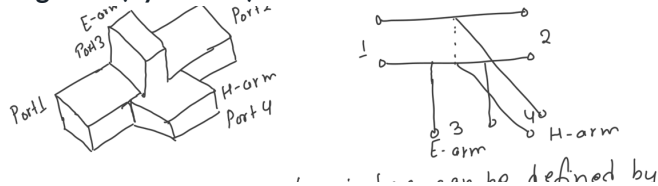


- Side arm in the plane of the **E** field, so it excites the collinear arms in **antiphase**.

$$[S] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- Fingerprint:** the $-1/\sqrt{2}$ in the third row / column.
- Use:** power splitting with a phase reversal; the difference arm of a magic tee.
- Ports 1 and 2 are not matched** ($S_{11} = S_{22} = 1/2$), which is the 3-port theorem of §3.5 showing up in the answer.

Magic tee (hybrid tee): the exam favourite

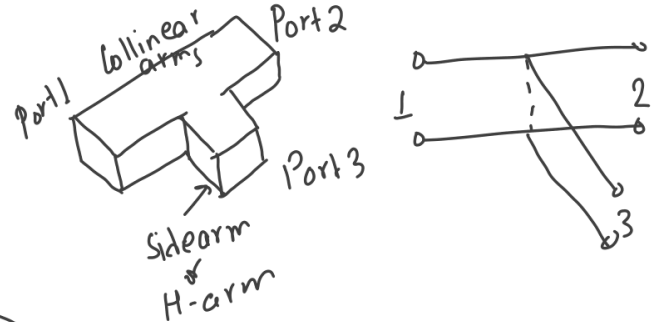


- Ports 1 and 2 collinear, port 3 the **E-arm** (difference), port 4 the **H-arm** (sum). Four physical facts:
 - E-arm drive gives antiphase outputs: $S_{23} = -S_{13}$.
 - H-arm drive gives in-phase outputs: $S_{24} = +S_{14}$.
 - E-arm and H-arm are decoupled: $S_{34} = S_{43} = 0$.
 - Ports 3 and 4 matched: $S_{33} = S_{44} = 0$; and the collinear arms are decoupled, $S_{12} = S_{21} = 0$.
- The unitary property then forces $S_{11} = S_{22} = 0$ as well, so the tee ends up matched at **all four ports**:

$$[S] = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

- Fingerprint:** an all-zero 2×2 block on each diagonal, one column with a sign flip (E-arm), one without (H-arm).

H-plane tee (shunt tee)



- Side arm in the plane of the **H** field, so the collinear arms come out **in phase**.

$$[S] = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

- Fingerprint:** the minus sign has moved to S_{12} ; the side-arm column is all positive.
- Use:** in-phase power divider or combiner; the sum arm of a magic tee.
- E-tee vs H-tee in one line:** same geometry, same magnitudes, opposite phase relation. The E-tee subtracts, the H-tee adds.

Circulator, isolator, gyrator

- Circulator:** $[S]$ as in §3.5, matched and lossless but **not** symmetric.
- Isolator:** a circulator with port 3 loaded, $[S] = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$; forward passes, reverse is absorbed.
- Gyrator:** $[S] = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, a 180° phase difference between the two directions.

Directional coupler

- Through arm α , coupled arm $j\beta$, isolated arm 0, with $\alpha^2 + \beta^2 = 1$:

$$[S] = \begin{bmatrix} 0 & \alpha & j\beta & 0 \\ \alpha & 0 & 0 & j\beta \\ j\beta & 0 & 0 & \alpha \\ 0 & j\beta & \alpha & 0 \end{bmatrix}$$

- A 3 dB coupler has $\alpha = \beta = 1/\sqrt{2}$, and is then a magic tee in quadrature form.

Duplexer

- **Circulator duplexer** (the usual answer): port 1 transmitter, port 2 antenna, port 3 receiver. Ideally lossless, so **all** the transmitter power reaches the antenna and **all** the echo reaches the receiver, with the transmitter isolated from the receiver.
- **Magic-tee duplexer**: transmitter on the H-arm, antenna and a matched dummy load on the collinear arms, receiver on the E-arm. Splits the transmitter power in **half** and keeps transmitter and receiver isolated. This is the version **76 Bh** asks for (§3.8).

verified/added **Magic tee with both E- and H-arms short-circuited** 4+4

PIN 1/24 82 Bh · not in any source note, derived here and checked in `sparam.py`

- Shorts at the junction reference plane give $a_3 = -b_3$ and $a_4 = -b_4$.
- Substituting into the magic-tee equations collapses the 4-port to the two-port seen from ports 1 and 2:

$$[S]_{\text{shorted}} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Amplifier, attenuator, amplitude modulator

- Matched amplifier: $[S] = \begin{bmatrix} 0 & S_{12} \\ S_{21} & 0 \end{bmatrix}$ with $|S_{21}| > 1 \gg |S_{12}|$. Not reciprocal, not lossless.
- Matched attenuator of L dB: $[S] = \begin{bmatrix} 0 & \alpha \\ \alpha & 0 \end{bmatrix}$, $\alpha = 10^{-L/20}$. Reciprocal but lossy.
- **Amplitude modulator (78 Ch)**: a matched, reciprocal two-port whose transmission is **driven by the modulating signal**,

$$[S] = \begin{bmatrix} 0 & \alpha(t) \\ \alpha(t) & 0 \end{bmatrix}, \quad \alpha(t) = \alpha_0 (1 + m \cos \omega_m t)$$

Problem 6 of the companion derives it and evaluates the numbers.

- **Behaviour**: each collinear arm reflects **all** of its incident power with a 180° phase reversal, and the two arms stay **completely isolated** from each other.
- The E-arm and H-arm reflections add in the driven arm and cancel in the other one, which is why S_{21} stays zero.
- Still lossless (each column sums to 1), still reciprocal. The tee has become a **pair of independent short circuits**, useful as a reflective phase shifter or a reference short.

3.7 Identifying a Device from its S-Matrix

Identify and explain, with a neat diagram, the passive microwave device described by the given S-matrix **5 6 8 10**

HOT 5/24 74 Bh, 75 Bh, 78 Ch, 81 Bh, 82 Bh · in all five papers the printed matrix is a magic tee

The six-step method

- Count the ports:** the order of the matrix is the port count. 4×4 narrows it to coupler, magic tee or hybrid ring immediately.
- Read the diagonal:** zeros mark the matched ports. An all-zero diagonal on a 4-port means matched everywhere.
- Test the transpose:** symmetric means passive and isotropic; asymmetric means ferrite or active.
- Test the columns:** sums of 1 and orthogonal pairs mean lossless. A shortfall is dissipation.
- Map the zeros:** an off-diagonal zero is an **isolated pair**. Which pairs are decoupled is the strongest single clue.
- Read the signs and the j 's:** a sign flip down a column means an **antiphase (difference, E) arm**; matching signs mean an **in-phase (sum, H) arm**; a factor j means a **quadrature coupled arm**.

Answer skeleton

- Name the device in the first line. Do not narrate the tests first.
- Draw it: the 3D waveguide sketch with all arms labelled, plus the schematic.
- Justify with the four tests, one line each.
- Close with two uses and one limitation (the tee needs matching irises at the collinear arms to reach the ideal matrix in practice).

The four matrices the papers actually print

- 78 Ch 10**, printed without its normalising factor:

$$[S] = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 1 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

Zero diagonal, symmetric, the (1, 2) and (3, 4) blocks zero, column 3 sign-flipped: **magic tee**. It is unitary only after dividing by $\sqrt{2}$, and saying so earns the “judge the condition” mark.

- 81 Bh 5** and **74 Bh 8**, printed symbolically:

$$[S] = \begin{bmatrix} S_{11} & S_{12} & S_{13} & S_{14} \\ S_{12} & S_{22} & S_{13} & -S_{14} \\ S_{13} & S_{13} & 0 & 0 \\ S_{14} & -S_{14} & 0 & 0 \end{bmatrix}$$

Column 3 in phase (H-arm), column 4 antiphase (E-arm), $S_{33} = S_{44} = 0$: the **magic tee before the unitary property is applied**. Finish the job and $S_{11} = S_{22} = S_{12} = 0$, $S_{13} = S_{14} = 1/\sqrt{2}$.

Fingerprint table

Pattern	Device
2×2 , $S_{11} = S_{22} = 0$, $S_{12} = S_{21} = 1$	Matched lossless line or ideal matched network
2×2 , $S_{12} = 0$, $S_{21} = 1$	Isolator
2×2 , $ S_{21} > 1$, $ S_{12} $ tiny	Amplifier (active, not reciprocal)
3×3 , cyclic ones, not symmetric	Circulator
3×3 with a $\pm 1/\sqrt{2}$ column	E-plane tee (–) or H-plane tee (+)
4×4 , zero diagonal, two zero 2×2 blocks, one column sign-flipped	Magic tee
4×4 , zero diagonal, j on the coupled arm, one zero per row	Directional coupler

- 75 Bh 6**, the same device with the collinear isolation already written in:

$$[S] = \begin{bmatrix} S_{11} & 0 & S_{13} & S_{14} \\ 0 & S_{22} & -S_{13} & S_{14} \\ S_{13} & -S_{13} & 0 & 0 \\ S_{14} & S_{14} & 0 & 0 \end{bmatrix}$$

Here column 3 is the antiphase E-arm and column 4 the in-phase H-arm, the mirror of the previous labelling.

Read the signs, not the port numbers.

- 82 Bh 4** prints two *amplifier* sets instead, so steps 3 and 4 answer it: $S_{12} \neq S_{21}$ and $|S_{21}| > 1$ make both of them active and neither reciprocal (§3.4, Problem 2).

3.8 Power Splitting and Combining: the Radar Questions

Two identical radar transmitters must deliver twice the power to one antenna; and an ASR needs half the transmitter power at the antenna through a duplexer. Solve both with an S-matrix 6 8 10

HOT 4/24 73 Ma, 73 Bh, 74 Ma, 76 Bh · one device, two directions: the magic tee combines and splits

Combining: two transmitters into one antenna

- **Device:** magic tee (or a 3 dB hybrid coupler).
- **Connection:** the two transmitters drive the collinear arms 1 and 2 **in phase and at equal amplitude**; the antenna hangs on the H-arm (port 4); the E-arm (port 3) carries a **matched dummy load**.
- **Mathematics**, with $a_1 = a_2 = a$ and $a_3 = a_4 = 0$:

$$b_4 = \frac{1}{\sqrt{2}}(a_1 + a_2) = \sqrt{2}a \Rightarrow |b_4|^2 = 2|a|^2$$

$$b_3 = \frac{1}{\sqrt{2}}(a_1 - a_2) = 0$$

- **Result:** the antenna receives $2P$, twice what either transmitter can deliver, and the dummy load receives nothing.
- **Why the tee and not a plain T-junction:** $S_{12} = 0$, so the two transmitters are **isolated** and cannot pull or damage each other; and all four ports stay matched, so neither transmitter sees a VSWR.
- **Practical riders** worth a mark each:
 - Phase must be equalised, so the two feed lines are cut to the same electrical length.
 - Amplitude imbalance Δ shows up as $|b_3|^2 \neq 0$: the dummy load absorbs exactly the mismatch, which is what protects the transmitters.
 - The dummy load must be rated for the full unbalanced power.

Splitting: half the transmitter power to the antenna

- **Device:** the same magic tee, used as a duplexer, driven from the H-arm.
- **Connection:** transmitter on port 4 (H-arm), antenna on port 1, matched load on port 2, receiver on port 3 (E-arm).
- **Mathematics**, with $a_4 = a$ and all other inputs zero:

$$b_1 = \frac{1}{\sqrt{2}}a, \quad b_2 = \frac{1}{\sqrt{2}}a \Rightarrow |b_1|^2 = |b_2|^2 = \frac{1}{2}|a|^2$$

$$b_3 = 0$$

- **Result:** the antenna gets exactly $P/2$, as **76 Bh** requires, and the receiver on the E-arm gets **none** of the transmitter power, which is the whole point of a duplexer.
- **On receive**, the echo arrives at port 1 alone and divides between the E-arm and the H-arm, so half of it reaches the receiver. The 3 dB penalty in each direction is the price of the magic-tee duplexer, and naming it shows you understand the device rather than reciting it.
- **Better answer for a real ASR:** a **ferrite circulator** duplexer routes the full power $1 \rightarrow 2$ and the full echo $2 \rightarrow 3$ with no 3 dB loss. Quote it as the improvement whenever the question does not force the half-power split.

Answer skeleton for both

- Name the device, draw it with every arm labelled and every termination named.
- Write $[S]$ for the magic tee.
- Write the drive vector $[a]$ the connection creates.
- Multiply out, and read the answer off $|b|^2$.
- Close with the isolation and matching argument: that is the “sufficient explanation” the marks are for.

3.9 PYQ Mapping

Theory questions, covered in this document

- **Why S-parameters; importance in microwave network analysis; vs h-parameters** → §3.1. **The most repeated topic in the chapter, 13 papers.** 69 Bh 4, 70 Bh 4, 70 Ma 4, 80 Ch 5, 81 Bh 3
- **Derive S-parameters for a two-port network** → §3.2. 69 Bh 6, 70 Bh 4, 80 Ba 6, 80 Ch 5
- **Properties of the S-matrix** → §3.3. The four tests are also the identification tool of §3.7. 80 Bh 4
- **Return loss and insertion loss** → §3.4. 72 Ash 2, 79 Bh 1
- **Three-port network: analysis, properties, characteristic parameters** → §3.5, with the impossibility theorem and its three escapes. 72 Ash 5, 73 Ma 4, 73 Bh 4, 74 Ma 6, 82 Bh 4
- **Three-port parameters using an E-plane tee example** → §3.5 for the parameters, §3.6 for the matrix, companion Problem 5 for the derivation. 80 Bh 4
- **S-matrices of E-plane tee, H-plane tee and a duplexer, and their uses** → §3.6. 81 Ba 8

- **Magic tee with both E- and H-arms shorted** → §3.6, verified/added derived here because no source note carries it. 82 Bh 4
- **Identify the passive device from a printed S-matrix** → §3.7, with all four printed matrices worked. 74 Bh 8, 75 Bh 6, 78 Ch 10, 81 Bh 5
- **Radar power combining and duplexer power splitting** → §3.8. 73 Ma 8, 73 Bh 8, 74 Ma 6, 76 Bh 10
- **Evaluate the S-matrix of an amplitude modulator** → §3.6 and companion Problem 6. verified/added 78 Ch 7

Numerical content

- **This chapter has a numerical companion**, Ch3 - RF and MW Network Theory and Analysis - Numerical Problems and Solutions. It carries every S-parameter calculation the archive asks for: the 79 Bh reciprocal / lossless / return-loss / reflected-power question, the 82 Bh comparison of two amplifier sets, the four device-identification matrices, the E-plane, H-plane and magic-tee derivations, and the two radar power problems.
- **Not in this chapter**: amplifier stability (K , Δ , μ) and the unilateral / bilateral maximum gain calculations. Those ride on S-parameters but the archive files them under **RF Design Practices**, so they belong to Chapter 6. §3.4 gives only the reading of the numbers that 82 Bh asks for.

Syllabus coverage check

- *Scattering matrix and its properties* → §3.3, §3.5, §3.6 ✓
- *S-parameter derivation and analysis* → §3.1, §3.2, §3.4, §3.7, §3.8 ✓

Cross-references, so nothing is said twice

- **Chapter 1** §1.2 justifies S-parameter analysis as part of the lumped-vs-distributed answer. The full case is §3.1 here.
- **Chapter 2** §2.8 carries the two-line S-matrix rider tacked onto ten stub-tuning questions ($[\Gamma_L]$ before matching, $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ after). The theory behind it is §3.2 and §3.3.
- **Chapter 4** covers the physical construction of the tees, couplers, circulators and waveguide junctions whose matrices are catalogued in §3.6.

Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

Short	Means	Short	Means
RF	radio frequency	MW, M/W	microwave
EM, EMR	electromagnetic (radiation)	TEM	transverse electromagnetic mode
TE	transverse electric mode	TM	transverse magnetic mode
WG	waveguide	TL	transmission line
VSWR	voltage standing wave ratio	SWR	standing wave ratio
VNA	vector network analyzer	SA	spectrum analyzer
DUT	device under test	S-params	scattering parameters
RL	return loss	IL	insertion loss
ML	mismatch loss	NF	noise figure
BW	bandwidth	SNR	signal to noise ratio
Tx, TX	transmitter	Rx, RX	receiver
BTS	base transceiver station	GSM	global system for mobile communications
EIRP	effective isotropic radiated power	ERP	effective radiated power
SAR	specific absorption rate	SARPs	standards and recommended practices
HERP	hazards of EM radiation to personnel	RADAR	radio detection and ranging
ASR	airport surveillance radar	CW	continuous wave

Short	Means	Short	Means
TWT, TWTA	traveling wave tube (amplifier)	BWO	backward wave oscillator
CFA	cross-field amplifier	LNA	low noise amplifier
HPA	high power amplifier	SWS	slow wave structure
O-type	linear beam tube (axial B -field)	M-type	crossed-field tube ($E \perp B$)
MASER	microwave amplification by stimulated emission of radiation	LASER	light amplification by stimulated emission of radiation
MESFET	metal-semiconductor FET	HEMT	high electron mobility transistor
GaAs	gallium arsenide	FET	field effect transistor
BJT	bipolar junction transistor	PIN	p-type, intrinsic, n-type diode
IMPATT	impact ionization avalanche transit-time	TRAPATT	trapped plasma avalanche triggered transit
LPF	low pass filter	HPF	high pass filter
BPF	band pass filter	BSF	band stop filter
MAG	maximum available gain	MSG	maximum stable gain
MIC	microwave integrated circuit	MMIC	monolithic microwave integrated circuit
ADS	advanced design system	IF	intermediate frequency
LO	local oscillator	RFIC	radio frequency integrated circuit
PCB	printed circuit board	SMA	subminiature version A connector

Short	Means	Short	Means
attr(s)	attribute(s)	no.	number
freq	frequency	info	information
ops	operations	config	configuration
rm	remove	vs, Vs	versus
w/, w/o	with, without	Eg	example / for example
+Fig	draw a figure	(I)	importance / need / significance
→	leads to / therefore	+	and
=	is / equals	norm	normalized
max, min	maximum, minimum	calc	calculation / calculate
eq, eqn	equation	approx	approximately
pts	points	coeff	coefficient
temp	temperature	params	parameters

Standard and component terms (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ($1 \rightarrow 2 \rightarrow 3 \rightarrow 1$) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

Key mathematical symbols: Z_0 characteristic impedance · Y_0 characteristic admittance · Γ reflection coefficient · S scattering matrix · K Rollett's stability factor · Δ determinant of S-matrix · μ single parameter stability metric · G_T transducer gain · G_{TU} unilateral transducer gain · f_c cutoff frequency · λ_g guide wavelength · v_p phase velocity · v_g group velocity.

Exam-paper month codes (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

Bold year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

Tier chips over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth ≥ 6 marks.

Mark chips such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.